

Experiment: Implement Stochastic Gradient Descent with Momentum for parameter estimation and compare convergence behavior.

Problem Statement

Consider the following dataset for a simple linear regression problem:

$$X = [1, 2, 3, 4, 5, 6, 7, 8]$$

$$y = [5, 7, 9, 11, 13, 15, 17, 19]$$

The dataset follows the relationship

$$y = 2x + 3.$$

Assume the linear regression model:

$$\hat{y} = wx + b$$

where w is the weight and b is the bias.

The objective is to estimate the parameters w and b by minimizing the Mean Squared Error (MSE):

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2.$$

For a single training sample (x_i, y_i) , use the loss:

$$L_i = (y_i - \hat{y}_i)^2$$

Given Parameters

Parameter	Value
Initial weight w	0
Initial bias b	0
Initial velocity v_w	0
Initial velocity v_b	0
Learning rate η	0.001
Number of epochs	3000
Momentum coefficients β	0.0, 0.5, 0.9

Table 1: Parameters used for SGD with Momentum